

SOCIAL MEDIA SENTIMENT ANALYSIS DASHBOARD

PROJECT REPORT

1. Project Title

Social Media Sentiment and Brand Reputation Monitoring System using Python, PostgreSQL, and Power BI

2. Objective:

The objective of this project is to collect social media posts related to brands and products, analyze customer sentiments as positive, negative, or neutral, monitor brand reputation and customer feedback, identify trending hashtags and topics, compare performance across different social media platforms, and generate meaningful business insights to support better decision-making

3. Tools Used

Tool| Purpose

Microsoft Excel – Used to create the social media dataset, perform initial data cleaning, and organize the data.

Python – Used for data cleaning, sentiment analysis, text preprocessing, and libraries such as Pandas, NLTK, Matplotlib.pyplot, and Word Cloud.

PostgreSQL – Used for data storage, query processing, and reporting.

Power BI – Used for dashboard creation, KPI visualization, and business analysis.

4. Dataset Information

- Total Records : 900
- Platforms: Twitter, Facebook, Instagram, Reddit .
- Brands: Apple, Google, Vivo, OPPO, Samsung.
- Sentiments: Positive, Negative, Neutral.
- Metrics: Likes, Shares, Comments

5. Methodology

Data Preparation

- Created and cleaned the social media dataset in Excel.
- Used Python libraries such as Pandas, NLTK, Matplotlib, and WordCloud for preprocessing and analysis.
- Stored processed data in PostgreSQL.
- Connected the database to Power BI for dashboard creation and reporting.

6. Dashboard Pages and Visualizations

PAGE 1 - EXECUTIVE DASHBOARD

KPIs:

- Total Posts
- Positive Posts
- Negative Posts
- Neutral Posts
- Reputation Score

► Charts :

- Pie Chart – Percentage by Sentiment.
- Donut Chart – Percentage by Platform.

PAGE 2 - SENTIMENT ANALYSIS DASHBOARD

► Charts :

- Column Chart – Positive vs Negative vs Neutral
- Stacked Bar Chart – Platform-wise Sentiment
- Line Chart – Sentiment Trend

PAGE 3 - PLATFORM ANALYSIS DASHBOARD

Analysis:

- Sentiment Distribution by Platform
- Engagement Comparison by Platform
- Platform Performance Analysis
- Social Media Insights.

► Charts :

- Stacked Bar Chart
- Tree Map

PAGE 4 – TRENDING TOPICS DASHBOARD

Analysis:

- Top Trending Hashtags
- Hashtag Word Cloud
- Brand Mention Analysis
- Trend Insights

Page 5 – BRAND REPUTATION MONITORING DASHBOARD

Analysis:

- Reputation Score Trend
- Sentiment Breakdown by Brand
- Brand Filter
- Sentiment Analysis

Page 6 - ENGAGEMENT DASHBOARD

KPIs :

- Sum of Likes
- Sum of Shares
- Count of Comments

Analysis :

- Brand-wise Engagement Analysis.
- Platform Analysis.
- Engagement Insights.

7. Testing and Validation

- Verified sentiment classification results.
- Tested dashboard filters and interactions.
- Validated KPI calculations and visualizations.
- Ensured data consistency across platforms.

8. Challenges Faced

- Handling social media data from multiple platforms.
- Cleaning unstructured text data.
- Identifying accurate sentiment categories.
- Creating meaningful reputation metrics.
- Building interactive dashboards with multiple visualizations.

9. Conclusion

The project successfully analyzed social media data to monitor customer sentiment and brand reputation across multiple platforms. The dashboards provide valuable insights into customer opinions, brand perception, trending topics, and engagement patterns, helping organizations make data-driven decisions and improve customer satisfaction.

10. Future Scope

- Integrate real-time data using social media APIs.
- Implement advanced Machine Learning and NLP models.
- Develop automated alerts for negative sentiment spikes.
- Add predictive analytics for future sentiment trends.
- Deploy the solution on cloud platforms for real-time monitoring and enterprise usage.